COURIER MANAGEMENT SYSTEM (CMS)

Application : COURIER MANAGEMENT

Project Documentation

TEAM MEMBERS :

Team ID	:	SWTID-2026-2008
Team Size	:	5
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Courier Management System (CMS)

User Story:

Courier and logistics companies often struggle to efficiently manage shipments and deliveries due to reliance on manual processes and disconnected systems. The absence of a centralized shipment tracking mechanism makes it difficult to monitor parcel movement in real time, resulting in delayed delivery updates and poor visibility for customers. Delivery agent assignments are performed manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations. Collectively, these challenges result in customer dissatisfaction, reduced productivity, operational inefficiencies, and ultimately revenue loss for the organization.

1. INTRODUCTION

1.1 Project Overview :

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

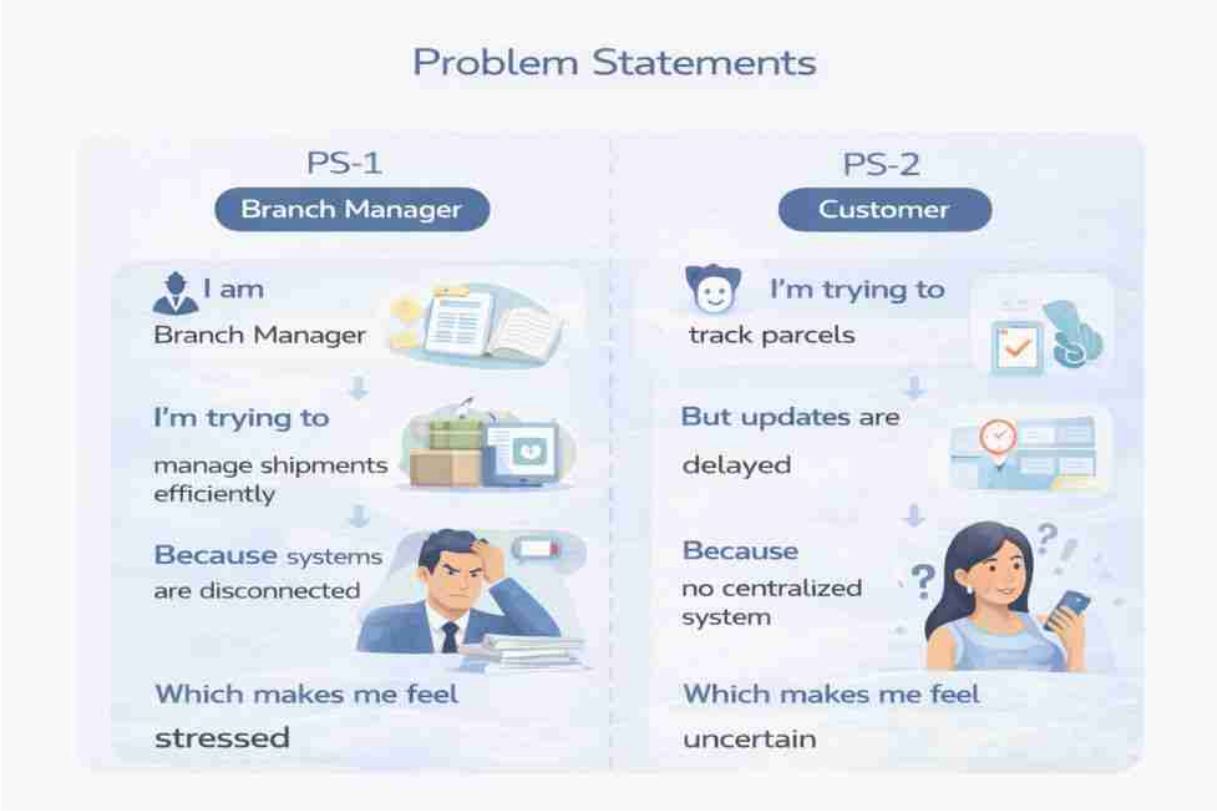
1.2 Purpose :

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

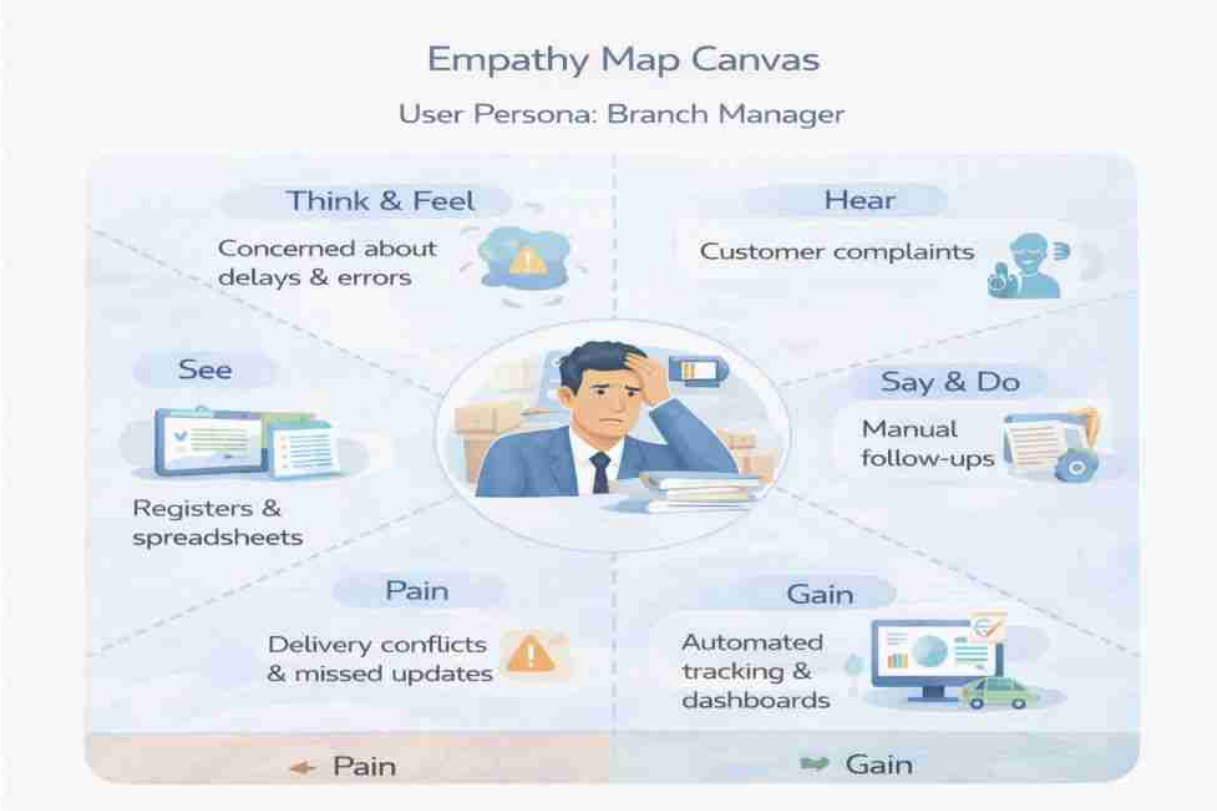
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnecte d	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain



2.2 Empathy Map Canvas

User Persona: Branch Manager

- **Think & Feel:** Concerned about delays & errors
- **Hear:** Customer complaints
- **See:** Registers & spreadsheets
- **Say & Do:** Manual follow-ups
- **Pain:** Delivery conflicts & missed updates
- **Gain:** Automated tracking & dashboards



2.3 Brainstorming & Idea Prioritization

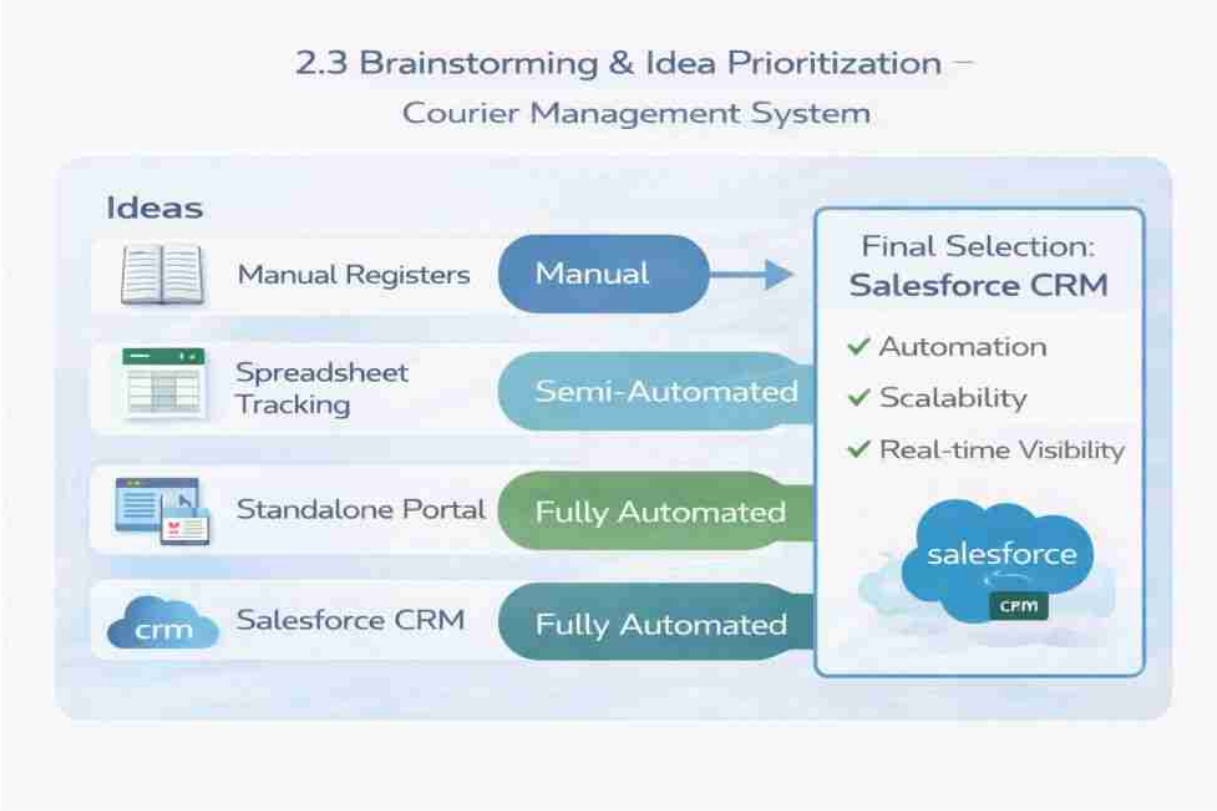
Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection

Salesforce CRM selected due to:

- Automation
- Scalability
- Real-time visibility



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation

FR-6	Reports & dashboards

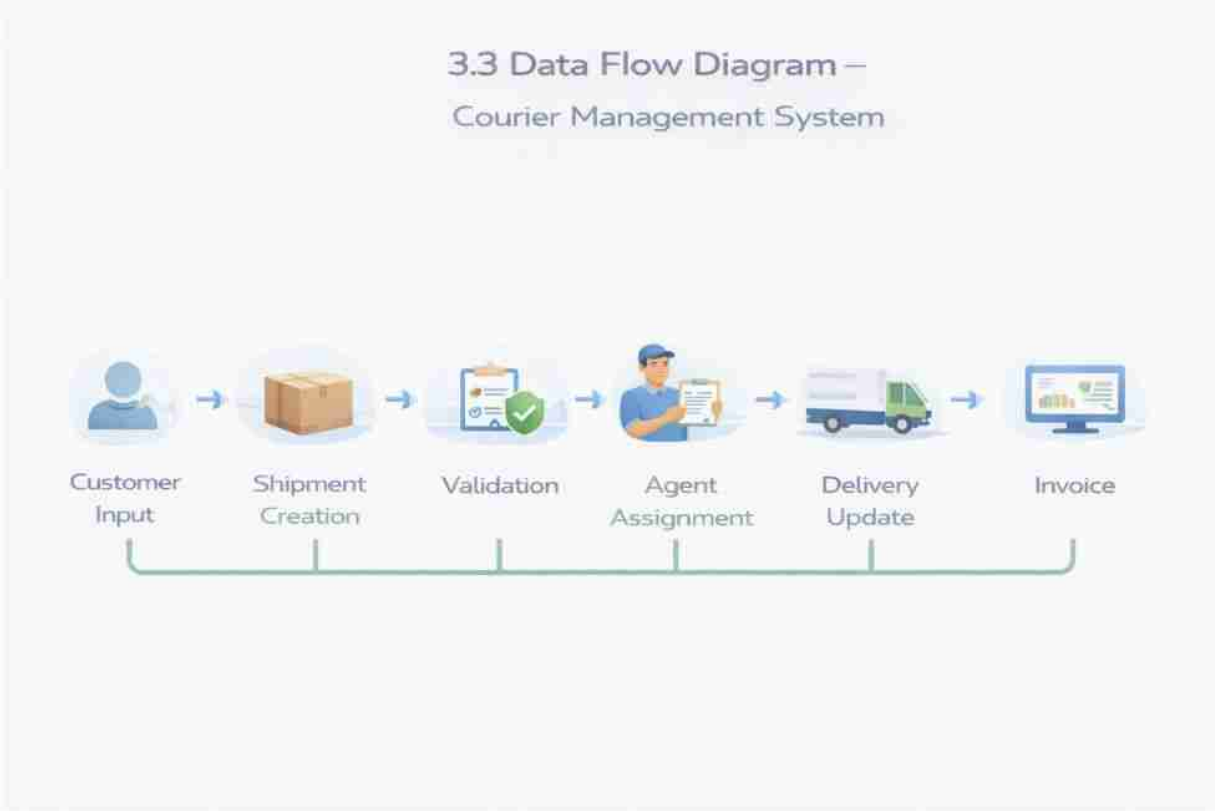
Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports



3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing
Impact	Faster deliveries
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery	5	High	Member2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			Agent, Branch, and Invoice objects with proper relationships so that courier data is stored and managed efficiently.			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place.

Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

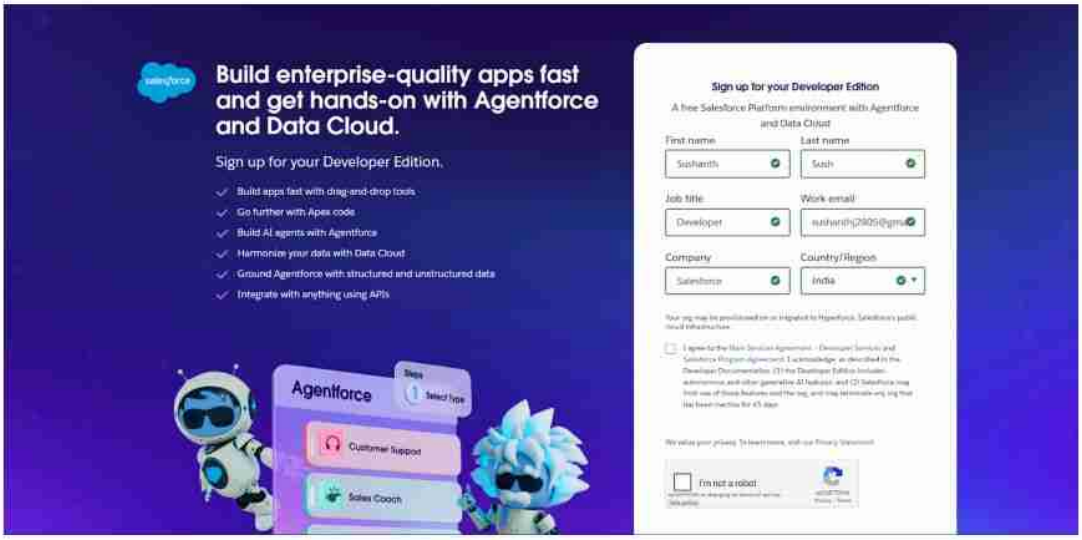
What Is Salesforce ?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email



1) First name & Last name

2) Email

3) Job Title: Developer

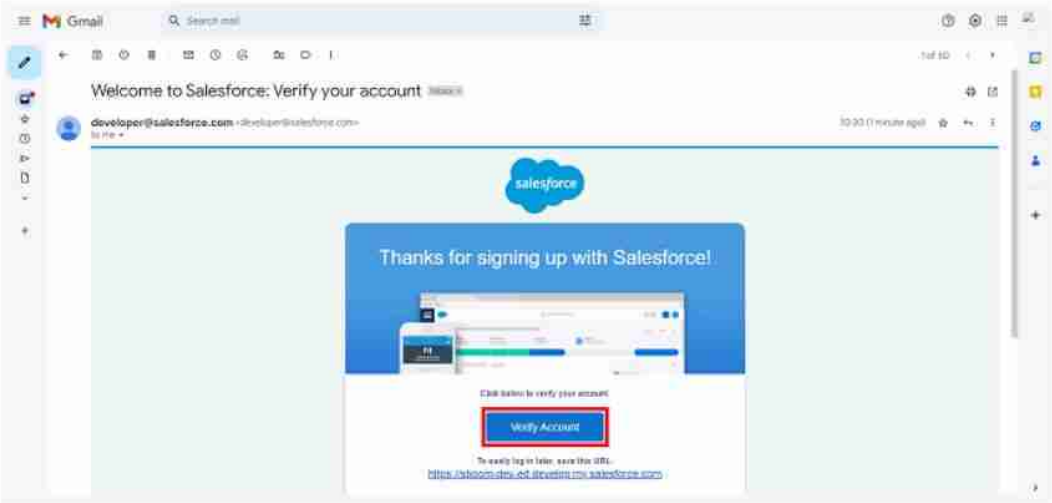
4) Company: College Name

5) Country: India

Click on sign me up after filling these.

Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account
3.Give a password and answer a security question and click on change password.
4.Then you will redirect to your salesforce setup page.

Milestone 2- Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Milestone 4: Fields & Relationships

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.
They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.
By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Milestone 8 - The Lightning App :

What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Milestone 9- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

Milestone 10- Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Milestone 11- User :

What is a user?

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

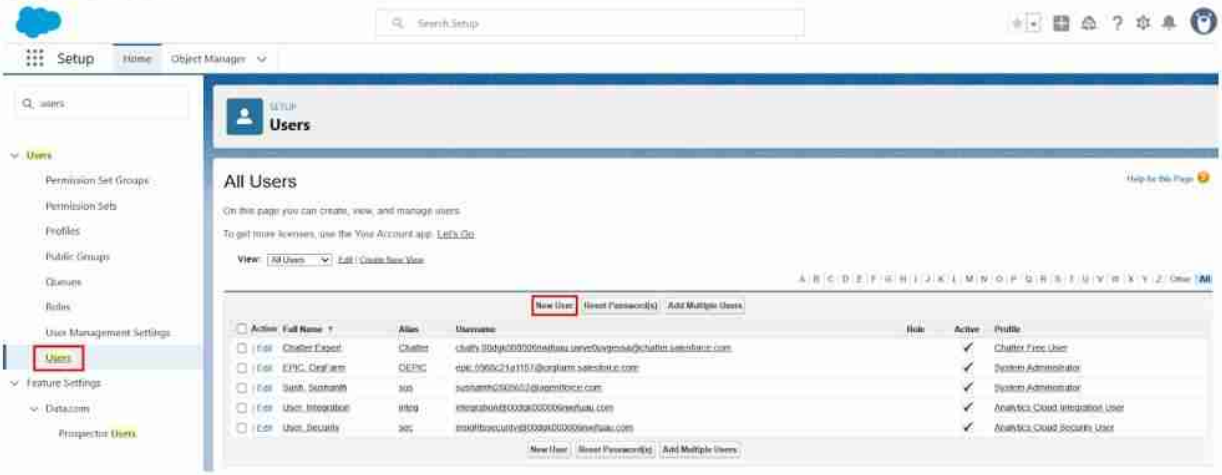
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.

2. Click New User.



Fill in the Fields:

Field Name	Value
First Name	Enter first name (Example: Ravi)
Last Name	Enter last name (Example: Kumar)

Alias	Give an alias name
Email ID	Enter your personal email ID
Username	Must be in the format: text@text.text and must be unique
Nickname	Give a nickname
Role	Select **Courier Admin**
User License	Salesforce
Profile	**Courier Admin**

3. Click Save.

Activity 2: Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support

Example:

To create a Branch Manager User:

- Go to Setup → Users → New User
- Fill details like:

Field	Example Value
First Name	Arjun
Last Name	Reddy
Email	arjun@gmail.com
Username	arjun.manager@cms.com
Role	Branch Manager
Profile	Branch Manager

- Click Save

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To Create Delivery Agent User:

Follow same steps with:

- Role: Delivery Agent
- Profile: Delivery Agent

To Create Customer Support User:

- Role: Customer Support
- Profile: Customer Support

Important Points

- Every user must have:
    - Unique username
    - Valid email
    - Assigned Role

    - Assigned Profile

- Profiles control permissions
- Roles control data visibility

Milestone 15 : Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.

Step 2: Create Shipment Report Type

- Primary Object: Shipment__c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports

- Deployment Status: Deployed
- Click Next → Save

Step 4: Create Agent Performance Report Type

- Primary Object: Delivery_Agent__c
- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Activity 2: Create Reports

Report 1: Shipment Status Summary Report

Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue

Step 2: Add Columns

Add these fields:

- Tracking Number
- Customer
- Status

- Expected Date
- Assigned Agent

Step 3: Add Filters

- Show Me: All Shipments
- Date Filter: This Month
- Status Filter: Select specific Status (Optional)

Step 4: Group & Summarize

- Group rows by Status__c
- Add summary on Price

Step 5: Save Report

- Click Save & Run
- Name: Shipment Status Summary
- Folder: CMS Reports

Report 2: Agent Performance Report

Step 1: Create Report

- New Report → Choose Agent Performance Report
- Click Continue

Step 2: Add Columns

- Agent Name
- Branch

- Number of Shipments
- Status

Step 3: Group Data

- Group by Delivery_Agent__c

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

Report 3: Revenue Report

Steps

- Create report using Shipment Tracking Report Type

Add fields:

- Tracking Number
- Customer
- Price
- Invoice Status

Group by:

- Branch
- Month

Save as: Revenue by Branch Report

- Chart Type: Bar Chart
- Click Add

Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Activity 5: Save & Activate Dashboard

1. Click Save
2. Click Done

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create

Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report

Final Layout :

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

Thank You